

The Quantization Effects of CORDIC Arithmetic for Digital Signal Processing Applications

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Abstract

CORDIC (COordinate Rotation DIgital Computer) is an algorithm for performing a sequence of iteration computation using the coordinate rotation. It can generate some powerful elementary function only realized by a simple set of adder and shifter. As such, it is suitable to implement high performance chips, which are applied in digital signal processing and image processing fields with high speed and massive function computation using VLSI technology.

In this paper, we discuss and analyze the computation error of the CORDIC algorithm. We analyze the CORDIC algorithm and categorize different types of computation error in order to derive from them systematically. Each type of error analysis includes rotation mode and vectoring mode according to characteristic of the CORDIC algorithm, and the computation error of each mode is made up of approximation error and truncation error. We have not only considered and promoted some neglected errors and formulas, but also revealed the outcomes of error analysis in the form, which is only related to the number of iteration and word length only. We also reveal the design reference with the error tolerance in tables.

Keywords: CORDIC, approximation error, truncation error, digital signal processing.

1. Introduction

CORDIC (COordinate Rotation DIgital Computer) is an iterative arithmetic algorithm for efficient computation of elementary functions [1]. Based on a rotation paradigm, CORDIC facilitates an efficient and unified computation scheme for the evaluation of many elementary functions such as trigonometric functions, hyperbolic functions, exponentials, logarithm, and square-roots [2]. This makes CORDIC particularly appealing to applications which require intensive evaluations of elementary functions.

Hardware implementation of CORDIC needs only shifters and adders, the simple hardware is very attractive. Haviland and Tuszynski [3] designed and implemented a full function CORDIC chip in 1980. T. Y. Sung and Y. H. Hu [4], [5] proposed doubly pipelined CORDIC array processors applying on digital signal processing and also implemented a CORDIC chip in 1987. In 1992, Y. H. Hu, H.E. Liao [6] proposed CORDIC algorithm applying on adaptive lattice filter. In 1985, T. Y. Sung and Y. H. Hu proposed CORDIC array processor applying on solving Toeplitz systems [7]. In 1987, T. Y. Sung and Y. H. Hu also proposed CORDIC arithmetic applying on Kalman filters [8]. S. S. Nikolaidis proposed CORDIC architecture applying on all-pass filters. J.

K. Hwang [9] proposed VLSI architecture of adaptive lattice filter basing on CORDIC, they improved the computation accuracy. In 1993, Metafas and Goutis [10] proposed floating-point computation architecture of CORDIC. X. Hu, Harber, and Bass [11] expanded convergence range of CORDIC and reduced input limits.

CORDIC is applied in signal processing widely today, but the research of computation error and accuracy analysis is rarely. Many designers of CORDIC algorithm might not understand the role of computation error in CORDIC design, they had just increased number of iteration, and even number of bits of mantissa is less considered. Error analysis of CORDIC algorithm was proposed by Y. H. Hu [12] who changed conventional design concept of CORDIC algorithm, but neglected an influence factor which was discussed about error of $\delta(i)$ sequence by X. Hu [13] and Kota [14]. Kota had just discussed in circular coordinate system, but the error analysis of CORDIC was not completely.

According to improving performance and saving hardware, the number of iteration and word length should be concerned. These concerns should cause computation error. Especially, the model of fixed point error analysis of the CORDIC algorithm would be proposed, and prediction of computation error with number of iteration and word length could be performed definitely. As knowing where the errors are, we can design the hardware with the error consideration in order to get the best cost-performance and the desired outcomes. An optimal design of CORDIC arithmetic for digital signal processing applications would be achieved.

2. The CORDIC Algorithm

Basic formula of CORDIC is as follows [2]:

$$x(i+1)=x(i)+m \delta(i)y(i)2^{-s(m,i)}, \quad (1)$$

$$y(i+1)=y(i)-\delta(i)x(i)2^{-s(m,i)}, \quad (2)$$

$$z(i+1)=z(i)-\delta(i)\alpha_m(i), \quad (3)$$

where $s(m,i)$ is a non-decreasing integer “shift sequence” satisfying

$$s(m,i) \leq s(m,i+1) \leq s(m,i)+1, \text{ and} \\ \alpha_m(i)=m^{-1/2} \tan^{-1}[m^{-1/2} 2^{-s(m,i)}] \quad (4)$$

is the angle rotated in i -th iteration. $m=+1, -1$ or 0 determines whether the coordinate system is circular, hyperbolic, or linear, respectively. $\delta(i)$ determines the direction of rotation angle, $+1$ and -1 represent different direction respectively.

3. Domain of Convergence of CORDIC

CORDIC is a numerical algorithm with iterations, the domain and input maximum angle of convergence must be concerned. Walther proposed converging criterion [2], but he had not proved it yet. Now, we prove this criterion and establish two theorems for domain of convergence.

Theorem 1: Converging criterion :

$$\alpha(i) - \sum_{j=i+1}^{n-1} \alpha(j) \leq \alpha(n-1), \text{ for } i \geq 0.$$

Proof: In case of linear coordinate system, rotation angle α is represented as

$$\alpha(i) = 2^{-i}, \quad \alpha(i+1) = 2^{-(i+1)},$$

$$\text{then } 2^{-i} - 2^{-(i+1)} = 2^{-(i+1)} \leq 2^{-(i+1)}.$$

$$\text{Hence, } \alpha(i) - \alpha(i+1) \leq \alpha(i+1).$$

In case of circular coordinate system, rotation angle $\alpha(i)$ and $\alpha(i+1)$ are represented as $\alpha(i) = \tan^{-1} 2^{-i} = \theta$ and $\alpha(i+1) = \tan^{-1} 2^{-(i+1)} = \varphi$ respectively.

$$\tan(\theta - \varphi) \leq \tan \varphi = \frac{\tan \theta - \tan \varphi}{1 + \tan \theta \tan \varphi}$$

$$= \frac{2^{-(i+1)}}{1+2^{-2i-1}} \leq 2^{-(i+1)}, \text{ and } \theta - \varphi \leq \varphi.$$

Hence, $\alpha(i) - \alpha(i+1) \leq \alpha(i+1)$.

In case of hyperbolic coordinate system, rotation angle $\alpha(i)$ and $\alpha(i+1)$ are represented as

$$\alpha(i) = \tanh^{-1} 2^{-i} = \theta \quad \text{and} \quad \alpha(i+1) = \tanh^{-1} 2^{-(i+1)} = \varphi$$

respectively.

$$\tanh^{-1} 2^{-i} - \tanh^{-1} 2^{-(i+1)} \leq \tanh^{-1} 2^{-(i+1)}.$$

Hence, $\alpha(i) - \alpha(i+1) \leq \alpha(i+1)$.

According to above derivations, we summarize as

$$\alpha(i) - \alpha(i+1) - \alpha(i+2) - \dots - \alpha(n-1) \leq \alpha(n-1).$$

Finally, the convergence criterion is represented as

$$\alpha(i) - \sum_{j=i+1}^{n-1} \alpha(j) \leq \alpha(n-1). \quad (5) \quad \square$$

Lemma 1: $|z_{\max}(0)| = \alpha(n-1) + \sum_{j=0}^{n-1} \alpha(j)$.

Proof: Based on convergence criterion, the rotation angle should be ranged from 0 to $\alpha(n-1)$.

$|z(0)|$ could be represented as

$$|z(0)| \leq \sum_{j=0}^{n-1} \alpha(j) + \alpha(n-1).$$

Since $z(0)$ can be represented as range of input

angle, $|z_{\max}(0)|$ can be represented as

$$|z_{\max}(0)| = \alpha(n-1) + \sum_{j=0}^{n-1} \alpha(j). \quad (6)$$

Hence, $|z_{\max}(0)|$ represents a maximum input angle

or a maximum convergence angle. $\square$

Theorem 2: $|z(i)| \leq \alpha(n-1) + \sum_{j=i}^{n-1} \alpha(j)$, for $i \geq 0$. (7)

Proof: According to theorem 1, we obtain

$$|z(i+1)| < \alpha(n-1) + \sum_{j=i+1}^{n-1} \alpha(j).$$

Hence, $|z(i)| \leq \alpha(n-1) + \sum_{j=i}^{n-1} \alpha(j)$. $\square$

Lemma 2: Convergence range can be expanded [11];

in linear, circular, and hyperbolic coordinate system,

$\theta_{\max}$ can be represented as $\theta_{\max} = 2^{M+1}$,

$$\theta_{\max} = \tan^{-1}(2^{-n}) + 2\tan^{-1}(2^0) + \sum_{j=0}^n \tan^{-1} 2^{-j}, \text{ and}$$

$$\theta_{\max} = \sum_{j=-M}^0 \tanh^{-1}(1 - 2^{-2^{j+1}}) + \tanh^{-1} 2^{-n} +$$

$$\sum_{j=1}^n \tanh^{-1}(2^{-j}), \text{ respectively.}$$

Proof: In linear coordinate system, $\frac{\alpha(i)}{\alpha(i+1)} \leq 2$

satisfies convergence criterion, such that

$$|z(0)| \leq \alpha(n) + \sum_{j=-M}^n \alpha(j). \quad (8)$$

Hence, $|z(0)|$ is expressed as

$$|z(0)| \leq \theta_{\max} = 2^{M+1}. \quad (9)$$

In circular coordinate system, let iteration sequence of j is 0, 0, 0, 1, 2, ..., n , and

$$\frac{\alpha(i)}{\alpha(i+1)} \leq 2 \text{ satisfies convergence criterion. The}$$

design of CORDIC hardware avoids left shift in order to simplify hardware, so that iteration sequence of j is chosen as 0, 0, 0, 1, 2, ..., n . $\theta_{\max}$ is represented as [8]

$$\begin{aligned} \theta_{\max} = & \tan^{-1}(2^{-n}) + 2\tan^{-1}(2^0) \\ & + \sum_{j=0}^n \tan^{-1} 2^{-j} \cong 3.3141(189.9^\circ) > 180^\circ. \end{aligned} \quad (10)$$

According to eq. (10), the convergence range is expanded to $\pm 180^\circ$. Hence, the range of input can be unlimited.

In case of hyperbolic coordinate system, when j is a negative number, $\tanh^{-1}(2^{-j})$ becomes a complex number, and the number of negative iteration should be reduced in order to improve computation speed [11]. Now,

$$\alpha(i) = \tanh^{-1}(1 - 2^{-2^{i+1}}), \text{ for } i \leq 0,$$

should be chosen.

$$\frac{\alpha(i)}{\alpha(i+1)} = \frac{\tanh^{-1}(1-2^{-2^{-i+1}})}{\tanh^{-1}(1-2^{-2^{-(i+1)+1}})} \leq 2, \text{ for } i \leq 0,$$

also satisfies convergence criterion, such that

$$\theta_{\max} = \sum_{j=-M}^0 \tanh^{-1}(1-2^{-2^{j+1}}) + \tanh^{-1}2^{-n} + \sum_{j=1}^n \tanh^{-1}(2^{-j}). \quad (11) \quad \square$$

In this paper, the angle α of hyperbolic coordinate system with expanding convergence range would be defined as

$$\alpha(i) = \tanh^{-1}(1-2^{-2^{-i+1}}), \text{ for } i \leq 0, \quad (12)$$

$$\alpha(i) = \tanh^{-1}(1-2^{-i}), \text{ for } i > 0. \quad (13)$$

4. Error Analysis of Conventional CORDIC Algorithm

The basic CORDIC algorithm is specified in eqs. (1), (2) and (3), vector $\mathbf{v}$ is defined as $[x \ y]^T$. Three kinds of representation form are defined [3]; one is a mathematical representation with infinite precision, as mathematically defined, values of this representation are defined as $\tilde{x}, \tilde{y}, \tilde{z}, \tilde{\mathbf{v}}, \theta$, another one is a hypothetical representation, every iteration uses the same $\delta(i)$ obtained in a real implementation and word length is performed with an infinite number of bits, values of this representation are defined as $x, y, z, \mathbf{v}, \alpha$, and the other one is a real-world representation, iteration and word length are performed both finite, values of this representation are defined as $\hat{x}, \hat{y}, \hat{z}, \hat{\mathbf{v}}, \varphi$.

In rotation mode, truncation error is represented as

$$\|\mathbf{v}(n) - \hat{\mathbf{v}}(n)\| \text{ which is a norm [3].}$$

$$\|\mathbf{v}(n) - \hat{\mathbf{v}}(n)\| \leq \|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| + \|\mathbf{v}(n) - \hat{\mathbf{v}}(n)\| \quad (14)$$

eq. (14) represents this overall approximation error [3].

In vectoring mode, truncation error is represented as $|\alpha - \varphi|$. Hence, this overall approximation error is represented as

$$|\theta - \varphi| \leq |\theta - \alpha| + |\alpha - \varphi|. \quad (15)$$

In case of rotation mode, absolute and relative approximation error are represented as

$$\|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| \quad \text{and} \quad \|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\|$$

respectively. Hence, results of error analysis only depend on “ n ” iterations, and b fractional binary digits, and “ b_v ” binary digits of mantissa of x, y . If n, b_v and b are determined, the result of error would be computed.

According to $\left| \theta - \sum_{i=0}^{n-1} \delta(i) \alpha(i) \right| < \alpha(n-1)$ [2], we

obtain

$$\left| \theta - \sum_{i=0}^{n-1} \delta(i) Q[\alpha(i)] \right| < Q[\alpha(n-1)], \quad (16)$$

where $Q[\bullet]$ denotes the operator of quantization, and $Q[x(i)]$ is defined as $Q[x(i)] = x(i) - e(i)$, where $e(i)$ is a truncation error and $e(i) < 2^{-b}$.

Hence,

$$\alpha - \sum_{i=0}^{n-1} \delta(i) Q[\alpha(i)] = \sum_{i=0}^{n-1} \delta(i) \mu(i),$$

where $\alpha = \sum_{i=0}^{n-1} \alpha(i)$, and $\mu(i) = \alpha(i) - Q[\alpha(i)]$. (17)

Assuming that μ is a maximum error for each $\mu(i)$, we obtain

$$\left| \alpha - \sum_{i=0}^{n-1} \delta(i) Q[\alpha(i)] \right| < n\mu. \quad (18)$$

Hence,

$$|\theta - \alpha| \leq Q[\alpha(n-1)] + n\mu. \quad (19)$$

Due to truncation error, $Q[\alpha(n-1)]$ is obtained as $Q[\alpha(n-1)] \leq \alpha(n-1)$, and $\mu = 2^{-b}$, where b is

fractional binary digits.

Theorem 3: In circular, hyperbolic and linear coordinate system, the relative approximation error of rotation mode are represented as

$$\begin{aligned} \|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| &\leq \tan^{-1} 2^{-(n+1)} + n \cdot 2^{-b}, \\ \|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| &\leq \left| e^{\tanh^{-1} 2^{-(n-r)} + n \cdot 2^{-b}} - 1 \right|, \end{aligned}$$

and $\|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| \leq 2^{-n} + n \cdot 2^{-b}$, respectively.

Proof: In circular coordinate system, $\mathbf{M}$ is defined as

$$\mathbf{M} = \begin{bmatrix} \cos \theta - \cos \alpha & \sin \theta - \sin \alpha \\ -(\sin \theta - \sin \alpha) & \cos \theta - \cos \alpha \end{bmatrix}.$$

Hence, maximum eigenvalue of $\mathbf{M}^T \mathbf{M}$ is $2 - 2\cos(\theta - \alpha)$, and $\|\mathbf{M}\| = \sqrt{2 - 2\cos(\theta - \alpha)}$.

$$\text{Hence, } \|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| \leq K_l \|\mathbf{v}(0)\| |\theta - \alpha|. \quad (20)$$

From eq. (19), $\|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\|$ should be modified

$$\text{as } \|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| \leq K_l \|\mathbf{v}(0)\| (Q[\alpha(n-1)] + n\mu).$$

According to eq. (20), we obtain

$$\|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| \leq \tan^{-1} 2^{-(n+1)} + n \cdot 2^{-b}. \quad (21)$$

In hyperbolic coordinate system,

$$\|\mathbf{M}\| = |\cosh \theta - \cosh \alpha + \sinh \theta - \sinh \alpha| = |e^{\theta} - e^{\alpha}|,$$

$$\text{and } \|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| = K_1 |e^{|\theta|} - e^{|\alpha|}| \|\mathbf{v}(0)\|. \quad (22)$$

$$\|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| \leq |e^{|\alpha| + |\theta|} - 1| \leq |e^{Q[\alpha(n-1)] + n\mu} - 1|.$$

$$\text{Hence, } \|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| \leq |e^{\tanh^{-1} 2^{-(n-r)} + n \cdot 2^{-b}} - 1|, \quad (23)$$

where r is the number of repeated iteration.

In linear coordinate system,

$$\begin{aligned} \|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| &= \left\| \begin{bmatrix} 0 & 0 \\ -(\theta - \alpha) & 0 \end{bmatrix} \right\| \|\mathbf{v}(0)\| \\ &= |\theta - \alpha| \|\mathbf{v}(0)\|. \end{aligned} \quad (24)$$

$\|\tilde{\mathbf{v}}(n)\|$ can be represented as

$$\|\tilde{\mathbf{v}}(n)\| = \left[\frac{\theta^2 + \sqrt{\theta^4 + 4\theta^2} + 2}{2} \right]^{\frac{1}{2}} \|\mathbf{v}(0)\|. \quad (25)$$

From eqs. (24) and (25), we obtain

$$0.29 |\theta - \alpha| < \|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| \leq |\theta - \alpha|, \quad (26)$$

where $-\pi \leq \theta \leq \pi$.

Finally, we obtain

$$\|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| \leq 2^{-n} + n \cdot 2^{-b}. \quad (27) \quad \square$$

Theorem 4: In circular, hyperbolic, linear coordinate system, the fixed point truncation error of rotation mode are represented

$$\text{as } \|Q[\hat{\mathbf{v}}(n)] - \mathbf{v}(n)\| \leq \sqrt{2}\varepsilon \left[1 + \sum_{j=1}^{n-1} \left(\prod_{i=j}^{n-1} \sqrt{1 + 2^{-2i}} \right) \right],$$

$$\|Q[\hat{\mathbf{v}}(n)] - \mathbf{v}(n)\| \leq \sqrt{2}\varepsilon \left\{ 1 + \sum_{j=1}^{n-1} \left[\prod_{i=j}^{n-1} \sqrt{1 + 2^{-2i}} \right] \cdot e^{\sum_{i=j}^{n-1} \tanh^{-1} 2^{-i}} \right\},$$

$$\text{and } |Q[\hat{y}(n)] - y(n)| = \sum_{i=2}^{n+1} |e(i)| = n\varepsilon, \text{ respectively.}$$

Proof: In circular coordinate system, the iteration process relates the output vector $\mathbf{v}(i+1)$ to its input vector $\mathbf{v}(i)$ as $\mathbf{v}(i+1) = \mathbf{p}_1(i)\mathbf{v}(i)$, where $\mathbf{p}_1(i)$ is represented the rotation in i -th iteration as

$$\mathbf{p}_1(i) = \begin{bmatrix} 1 & \delta(i)2^{-i} \\ -\delta(i)2^{-i} & 1 \end{bmatrix}.$$

Hence,

$$\|Q[\hat{\mathbf{v}}(n)] - \mathbf{v}(n)\| \leq |e(n)| + \sum_{j=1}^{n-1} \left\| \prod_{i=j}^{n-1} \mathbf{p}_1(i) \right\| \cdot |e(j)|. \quad (28)$$

$$\|Q[\tilde{\mathbf{v}}(n)] - \mathbf{v}(n)\| \leq \sqrt{2}\varepsilon \left[1 + \sum_{j=1}^{n-1} \left(\prod_{i=j}^{n-1} \sqrt{1+2^{-2i}} \right) \right], \quad (29)$$

where $|e(i)| = \sqrt{e_x^2(i) + e_y^2(i)} \leq \sqrt{2}\varepsilon$.

In hyperbolic coordinate system, from eq. (28), $Q[\tilde{\mathbf{v}}(n)] - \mathbf{v}(n)$ is represented as

$$\|Q[\tilde{\mathbf{v}}(n)] - \mathbf{v}(n)\| \leq \sqrt{2}\varepsilon \left\{ 1 + \sum_{j=1}^{n-1} \left[\prod_{i=j}^{n-1} \sqrt{1-2^{-2i}} \right] \cdot e^{\sum_{i=j}^{n-1} \tan^{-1} 2^{-i}} \right\}. \quad (30)$$

In linear coordinate system,

$Q[\tilde{y}(i+1)] = Q[\tilde{y}(i)] - \delta(i)\alpha(i)Q[\tilde{x}(i)] - e(i+1)$, and

$$Q[\tilde{y}(n)] = y(0) - \alpha \cdot x(0) - \sum_{i=1}^n e(i). \quad (31)$$

$$\text{Hence, } |Q[\tilde{y}(n)] - y(n)| = \sum_{i=2}^{n+1} |e(i)| = n\varepsilon. \quad (32) \quad \square$$

Theorem 5: In circular, hyperbolic and linear coordinate system, the approximation error of vectoring mode are represented

as $|\theta - \alpha| < \sin^{-1}(Q[\alpha(n-1)] + n\mu)$,

$$|\theta - \alpha| \leq \sinh^{-1} \left| e^{|\alpha|} - e^{|\theta|} \right| \leq \sinh^{-1} \left| e^{|\alpha-\theta|} - 1 \right|, \text{ and}$$

$$|\theta - \alpha| \leq Q[\alpha(n)] + n\mu, \text{ respectively.}$$

Proof: In circular coordinate system, using eq. (20) and $\tilde{\mathbf{y}}(n) = 0$, we obtain

$$|\theta - \alpha| < \sin^{-1}(Q[\alpha(n-1)] + n\mu). \quad (33)$$

In hyperbolic coordinate system, using eq. (22) and $\tilde{\mathbf{y}}(n) = 0$, we obtain

$$|\theta - \alpha| \leq \sinh^{-1} \left| e^{|\alpha|} - e^{|\theta|} \right| \leq \sinh^{-1} \left| e^{|\alpha-\theta|} - 1 \right|, \quad (34)$$

In linear coordinate system, from eqs. (18), (24) and $\tilde{\mathbf{y}}(n) = 0$, we obtain

$$|y(n)| \leq |x(0)|(Q[\alpha(n-1)] + n\mu). \quad (35)$$

From eq. (35), and $|\theta - \alpha| = \left| \frac{y(n)}{x(0)} \right|$, we obtain

$$|\theta - \alpha| \leq Q[\alpha(n)] + n\mu. \quad (36) \quad \square$$

Theorem 6: In circular, hyperbolic and linear coordinate system, the fixed point truncation error of

vectoring mode are represented as

$$|\alpha - \phi| = \left| \alpha - \sum_{i=0}^{n-1} \delta(i)Q[\alpha(i)] \right| \leq n\mu, \text{ where } \mu = 2^{-b}.$$

Proof: In circular coordinate system, from eqs. (17) and (18), we obtain

$$|\alpha - \phi| = \left| \alpha - \sum_{i=0}^{n-1} \delta(i)Q[\alpha(i)] \right| \leq n\mu, \text{ where } \mu = 2^{-b}. \quad (37)$$

Truncation errors of hyperbolic and linear coordinate system are the same as eq. (37). $\square$

5. Error Analysis of CORDIC Algorithm with Expanding Convergence Range

The CORDIC algorithm with expanding convergence range had been proposed [11], the error analysis of linear, circular and hyperbolic coordinate system is discussed in order.

In case of linear coordinate system, the sequence of iteration is $-M, -M+1, \dots, 0, 1, 2, \dots, n$. Convergence range is expanded; $|z_{in}| \leq 1$ is replaced by $|z_{in}| \leq 2^{M+1}$, $|y_{in}/x_{in}| \leq 1$ is replaced by $|y_{in}/x_{in}| \leq 2^{M+1}$, z is an integer of 2^{M+1} , b is replaced by $b-M-1$, the truncation error with expanding convergence range is 2^{M+1} times of that without expanding convergence range. Besides, N -iteration is replaced by $(N+M+1)$ -iteration.

Lemma 3: In linear, circular, and hyperbolic coordinate system, the relative approximation error of rotation mode are represented as

$$\|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| \leq Q[\alpha(n)] + (N+M+1) \cdot 2^{-b+M+1},$$

$$\|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| < (Q[\alpha(n-1)] + (N+2) \cdot 2^{-b+1}),$$

$$\text{and } \|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| \leq \left| e^{Q[\alpha(n-r)] + (N+M+1)\mu} - 1 \right|,$$

respectively.

Proof: In linear coordinate system, from *theorem 3*, n is replaced by $(N + M + 1)$, and 2^{-b} is replaced by 2^{-b+M+1} , we obtain

$$\|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| \leq Q[\alpha(n)] + (N+M+1) \cdot 2^{-b+M+1}. \quad (38)$$

In circular coordinate system, the convergence angle exceeds 180° , it makes input range unlimited, the sequence of iteration is $0, 0, 0, 1, 2, \dots, N$. Hence,

$$|z_{in}| \leq 3.3141 (189.9^\circ), \text{ and } K_1' = \prod_{i=0,0,0}^N \sqrt{1+2^{-2i}} = 2K_1.$$

For fixed word length, scale factor with expanding convergence range is two times of that without expanding convergence range, it causes that truncation error with expanding convergence range becomes two times of that without expanding convergence range, such that $\varepsilon = 2^{-b+1}$, $z = 3.3143$.

Hence, $b' = b - 1$, $\mu = 2^{-b'} = 2^{-b+1}$.

From eq. (38), $n = N + 2$, $\mu = 2^{-b+1}$, we obtain

$$\|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| < (Q[\alpha(n-1)] + (N+2) \cdot 2^{-b+1}). \quad (39)$$

In hyperbolic coordinate system, the analysis of approximation error of rotation mode is discussed as follows:

According to eq. (11), $\theta_{\max} = 2.0911550745$ ($M=0$), 3.8081486768 ($M=1$), 6.9263334719 ($M=2$), 12.8180806919 ($M=3$), 24.25550091711 ($M=4$).

$\theta_{\max}$ becomes larger, the number of bits of mantissa is reduced, such that $b' = b - M$, the truncation error with expanding convergence range becomes 2^M times of that of conventional CORDIC. Scale factor

$$K_h = \left[\prod_{i=-M}^0 \sqrt{1 - (1 - 2^{-2^{i+1}})^2} \right] \left[\prod_{i=1}^N \sqrt{1 - 2^{-2i}} \right].$$

$$K_h = 0.6614378278 < 2^0 \text{ } (M=0), 0.2301706229 < 2^{-2} \text{ } (M=1), 0.0203245237 < 2^{-5} \text{ } (M=2), 0.0001122778 < 2^{-13} \text{ } (M=3), 0.0000000024 < 2^{-28} \text{ } (M=4).$$

If $M=2$, number of bits of mantissa would be lose 5-bit. Hence, M is larger; the truncation error becomes much larger. Because of $b' < b$, $\varepsilon = 2^{-b'}$ becomes larger.

The sequence of iteration is $-M, -M+1, \dots, 0, 1, 2, 3, \dots, N$. n is replaced by $(N+M+1)$

and μ is represented as $\mu = 2^{-b'}$.

From *Theorem 3*, we obtain

$$\|\tilde{\mathbf{v}}(n) - \mathbf{v}(n)\| / \|\tilde{\mathbf{v}}(n)\| \leq \left| e^{Q[\alpha(n-r)] + (N+M+1)\mu} - 1 \right|, \quad (40)$$

where $\mu = 2^{-b'}$ and $b' = b - M$. $\square$

Lemma 4: In linear, circular, and hyperbolic coordinate system, the fixed point truncation error of rotation mode is represented as $|Q[\hat{y}(n)] - y(n)| = (N + M + 1)\varepsilon$,

$$|Q[\hat{\mathbf{v}}(n)] - \mathbf{v}(n)| \leq \sqrt{2}\varepsilon \left[1 + \sum_{j=0,0,1}^{n-1} \left(\prod_{i=j}^{n-1} \sqrt{1+2^{-2i}} \right) \right],$$

$$|Q[\hat{\mathbf{v}}(n)] - \mathbf{v}(n)| \leq \sqrt{2}\varepsilon \left\{ 1 + \sum_{j=-M+1}^n \left[\prod_{i=j}^n \sqrt{1-2^{-2i}} \right] \cdot e^{\sum_{i=j}^n \alpha(i)} \right\},$$

respectively.

Proof: In linear coordinate system, from *Theorem 4*, $Q[\hat{y}(n)]$ is represented

$$\text{as } Q[\hat{y}(n)] = y(-M) - \alpha_M \cdot x(-M) - \sum_{i=-M}^n e(i).$$

$$\text{Hence, } Q[\hat{y}(n)] = y(n) - \sum_{i=-M}^n e(i).$$

Assume that $e(i) = \varepsilon$, $|Q[\hat{y}(n)] - y(n)|$ is represented as $|Q[\hat{y}(n)] - y(n)| = (N + M + 1)\varepsilon$. (41)

In circular coordinate system, from *Theorem 4*, we obtain

$$\|Q[\tilde{\mathbf{v}}(n)] - \mathbf{v}(n)\| \leq \sqrt{2}\varepsilon \left[1 + \sum_{j=0,0,1}^{n-1} \left(\prod_{i=j}^{n-1} \sqrt{1+2^{-2i}} \right) \right]. \quad (42)$$

In hyperbolic coordinate system, from *Theorem 4*, $\|Q[\tilde{\mathbf{v}}(n)] - \mathbf{v}(n)\|$ is represented as

$$\|Q[\tilde{\mathbf{v}}(n)] - \mathbf{v}(n)\| \leq \sqrt{2}\varepsilon \left\{ 1 + \sum_{j=-M+1}^n \left[\prod_{i=j}^n \sqrt{1-2^{-2i}} \right] \cdot e^{\sum_{i=j}^n \alpha(i)} \right\}. \quad (43) \square$$

Lemma 5: In linear coordinate system, the approximation error of vectoring mode is represented as $|\theta - \alpha| = Q[\alpha(n)] + (N + M + 1) \cdot 2^{-b+M+1}$, $|\theta - \alpha| < \sin^{-1}(Q[\alpha(n-1)] + (N + 2) \cdot 2^{-b+1})$, and $|\theta - \alpha| \leq \sinh^{-1}(e^{Q[\alpha(n-r)] + (N+M+1)\mu'} - 1)$,

where $\mu' = 2^{-b'}$, $b' = b - M$, respectively.

Proof: In linear coordinate system, from *Theorem 5*, $|\theta - \alpha| = Q[\alpha(n)] + n\mu$, n is replaced by $(N + M + 1)$, and 2^{-b} is replaced by 2^{-b+M+1} , we obtain $|\theta - \alpha| = Q[\alpha(n)] + (N + M + 1) \cdot 2^{-b+M+1}$. (44)

In circular coordinate system, from *Theorem 5* and *Lemma 3*, $|\theta - \alpha|$ is represented as $|\theta - \alpha| < \sin^{-1}(Q[\alpha(n-1)] + (N + 2) \cdot 2^{-b+1})$. (45)

In hyperbolic coordinate system, according to *Theorem 5*, and expanding convergence range, $|\theta - \alpha|$ is represented as

$$|\theta - \alpha| \leq \sinh^{-1}(e^{Q[\alpha(n-r)] + (N+M+1)\mu'} - 1), \quad (46)$$

where $\mu' = 2^{-b'}$, $b' = b - M$. □

Lemma 6: In linear, circular and hyperbolic coordinate system, the fixed point truncation error of vectoring mode are represented

as $|\alpha - \phi| \leq (N + M + 1)\mu_M$, where $\mu_M = 2^{-b'}$,

where $b' = b - (M + 1)$, $|\alpha - \phi| \leq (N + 2)\mu_0$,

where $\mu_0 = 2^{-b'}$, $b' = b - 1$, and

$$|\theta - \alpha| \leq (N + M + 1)\mu.$$

Proof: In linear coordinate system, $|\alpha - \phi|$ is

$$\text{represented as } |\alpha - \phi| = \left| \sum_{i=-M}^n \delta(i)\mu(i) \right| \leq \sum_{i=-M}^n \mu(i).$$

Assume that $\mu(i) = \mu_M$, one has

$$|\alpha - \phi| \leq (N + M + 1)\mu_M, \quad \text{where } \mu_M = 2^{-b'},$$

where $b' = b - (M + 1)$.

In circular coordinate system, the sequence of iteration is changed, such that $|\alpha - \phi| \leq \sum_{i=0,0,0,1}^{n-1} \mu(i)$.

Assume that $\mu(i) = \mu_0$, one has

$$|\alpha - \phi| \leq (N + 2)\mu_0, \quad (47)$$

where $\mu_0 = 2^{-b'}$, $b' = b - 1$.

In hyperbolic coordinate system, due to expanding convergence range, $|\theta - \alpha|$ is represented as

$$|\theta - \alpha| \leq \left| \sum_{i=-M}^n \delta(i)\mu(i) \right| \leq \sum_{i=-M}^n \mu(i),$$

where $\mu' = 2^{-b'}$, $b' = b - M$.

$$|\theta - \alpha| \text{ can be represented as } |\theta - \alpha| \leq (N + M + 1)\mu. \quad (48) \square$$

6. Summary and Conclusion

The approximation error of rotation mode is represented relative error model; this error is related to iterations and word length. The truncation error is not related to initial values; hence this error is represented absolute error model. If the overall error is represented as relative error model, the truncation error is divided by ideal value $\|\tilde{\mathbf{v}}(n)\|$. If the overall error is represented as absolute error model, the approximation error of rotation mode is multiplied by

ideal value $\|\tilde{v}(n)\|$.

In this paper, we analyze the computation error of CORDIC for different type, and derive error analysis formulas which consider error of $\delta(i)$ sequence and expanding convergence range. The result of overall error for each type is shown in Tables 1 and 2. Comparison of Sung, Y. H. Hu (Hu) [12], and K. Kota (Kota) [14] research is shown in Table 3.

The computation error of CORDIC with expanding convergence range is discussed and analyzed, and error analysis formulas are also

developed. We establish complete error model involving error of $\delta(i)$ sequence [12], which Kota had just discussed in circular coordinate system [14]. All of error analysis formulas, which consist of truncation error and approximation error, are only related to word length and iteration number. According to analysis formulas and design reference tables, we can develop a cost-effective special purpose processor with CORDIC. Small area and high performance of CORDIC processor will be achieved.

Linear Coordinate System:
a. Rotation Mode: $[2^{-n} + n \cdot 2^{-b}] + n \cdot 2^{-b_v}$
b. Vectoring Mode: $[2^{-n} + n \cdot 2^{-b}] + n \cdot 2^{-b}$
Circular Coordinate System:
a. Rotation Mode: $[\tan^{-1} 2^{-(n-1)} + n \cdot 2^{-b}] + \sqrt{2} \cdot 2^{-b_v} \left[1 + \sum_{j=1}^{n-1} \left(\prod_{i=j}^{n-1} \sqrt{1+2^{-2i}} \right) \right]$
b. Vectoring Mode: $[\sin^{-1} (\tan^{-1} 2^{-(n-1)} + n \cdot 2^{-b})] + n \cdot 2^{-b}$
Hyperbolic Coordinate System:
a. Rotation Mode: $\left[\left e^{\tanh^{-1} 2^{-(n-r)} + n 2^{-b}} - 1 \right + \sqrt{2} \cdot 2^{-b_v} \left\{ 1 + \sum_{j=2}^n \left[\prod_{i=j}^n \sqrt{1-2^{-2i}} \right] \cdot e^{\sum_{i=j}^n \tanh^{-1} 2^{-i}} \right\} \right]$
b. Vectoring Mode: $[\sinh^{-1} (e^{\tanh^{-1} 2^{-(n-r)} + n 2^{-b}} - 1)] + n \cdot 2^{-b}$

Table 1 Overall error of conventional CORDIC algorithm ($\|\tilde{v}(n)\| = 1$)

Linear Coordinate System:
a. Rotation Mode: $[2^{-n} + (n + M + 1) \cdot 2^{-b+M+1}] + (n + M + 1) \cdot 2^{-b_v}$
b. Vectoring Mode: $[2^{-n} + (n + M + 1) \cdot 2^{-b+M+1}] + (n + M + 1) \cdot 2^{-b+M+1}$
Circular Coordinate System:
a. Rotation Mode: $[\tan^{-1} 2^{-(n-1)} + (n + 2) \cdot 2^{-b+1}] + \sqrt{2} \cdot 2^{-b_v+1} \left[1 + \sum_{j=0,0,1}^{n-1} \left(\prod_{i=j}^{n-1} \sqrt{1+2^{-2i}} \right) \right]$
b. Vectoring Mode: $[\sin^{-1} (\tan^{-1} 2^{-(n-1)} + (n + 2) \cdot 2^{-b+1})] + (n + 2) \cdot 2^{-b+1}$
Hyperbolic Coordinate System:
a. Rotation Mode: $\left[\left e^{\tanh^{-1} 2^{-(n-r)} + (n+M+1) 2^{-b+M}} - 1 \right + \sqrt{2} \cdot 2^{-b_v'} \left\{ 1 + \sum_{j=M+1}^n \left[\prod_{i=j}^n \sqrt{1-2^{-2i}} \right] \cdot e^{\sum_{i=j}^n \tanh^{-1} 2^{-i}} \right\} \right]$
b. Vectoring Mode: $[\sinh^{-1} (e^{\tanh^{-1} 2^{-(n-r)} + (n+M+1) 2^{-b+M}} - 1)] + (n + M + 1) \cdot 2^{-b+M}$

Table 2 Overall error of CORDIC algorithm with expanding convergence range ($\|\tilde{v}(n)\| = 1$)

Type of error analysis of CORDIC	Sung	Hu[9]	Kota[11]
Conventional CORDIC Algorithm	Yes	*	@
CORDIC Algorithm with Expanding Convergence Range	Yes	No.	No.

Table 3 Comparison of related researches (*: Neglect error of $\delta(i)$ sequence. @: Only circular coordinate system)

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